

Assignment 5: Azure Prompt Flow & LangChain

Start Assignment

- Due Sunday by 11:59pm
- Points 5
- Submitting a file upload
- Available Mar 15 at 12am - Apr 5 at 11:59pm

Objective

Strengthen the performance of your AI application by applying flow engineering principles and using Azure Prompt Flow and LangChain to refine, automate, and optimize complex prompts. Through iterative experimentation, integrated workflows, and systematic evaluation, students learn to operationalize prompt engineering techniques in real, production-style LLM pipelines.

Research, Setup & Integration

Students begin by researching Azure Prompt Flow and LangChain capabilities, then setting up the environment and integrating their initial prompt into a working flow. This step ensures the app can run prompts programmatically, making it easier to test variations and automate experiments. The goal is to establish the technical foundation needed for flow-based prompt refinement.

Original Prompt Analysis

Students analyze their existing complex prompt to identify bottlenecks, weak reasoning paths, and opportunities for decomposition or chaining. Using principles of flow engineering, they map out where chain prompting, routing, or tool invocation can enhance clarity and consistency. This diagnostic step guides the design of their first refined flow.

Automation Tools

Students use Azure Prompt Flow and LangChain to run structured experiments across multiple prompt variants, parameter settings, and chain designs. These tools allow them to automate comparisons, visualize flow steps, and test how decomposition, routing, or helper prompts improve results. The focus is on leveraging frameworks to refine complex prompts more systematically than manual testing alone.

Iterative Improvement

Students conduct several iterations of prompt refinement, measuring model performance after each change. They document what they modified, why they modified it, and the quantitative or qualitative impact of each iteration. By the end, they produce a data-backed improvement history that shows how flow engineering increased performance.

Evaluation

Students evaluate their final, refined prompts using predefined metrics such as accuracy, coherence, groundedness, user satisfaction, or task-specific pass rates. They compare refined prompts against the original baseline to determine measurable improvement. To complete the evaluation, they gather peer or user feedback to validate real-world usefulness and clarity.